

Advanced Programming Practices

MTech 2026

Lab - 6

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP6_Qn.c (e.g. ABC2026001_APP6_Q1.c).
 - Every program must compile without warnings using the gcc compiler.
 - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
 - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
 - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
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1. Create a singly linked list by inserting nodes one by one such that the resulting linked list remains sorted in ascending order.
2. Write a program to insert a new node at a specified position in a singly linked list.
3. Write a program to search for a given value in a singly linked list.
4. Write a program to count the number of nodes in a singly linked list.
5. Write a C program to create a singly linked list with the number of nodes specified by the user.